Engineering Change Notice(ECN)

ECN-E-2606-07 (Flip Axis Chain Cable 설계 개선)

FA연구소 전장팀

작성	검토	승인
문민재	양원주	이태경
2026/06/05	2026/06/05	2026/06/05

NINEBELL

1. Engineering Change Request

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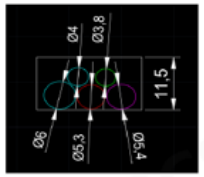
설계 변경 요청서(ECR)
(Engineering Change Request)

Customer	ACM
ECR No.	ECR-S-2605-03

Requesting department	Name / Department	HAOTIAN.MA/ME-1	변경 사유	☑ Customer Request ☐ Cost Reduction ☐ Quality improve ☐ Work Efficiency ☐ Discontinued ☐ 기타		
	Date	2026-04-30				
Request project	Category	Project Type	변경 Level (Change Level)	☐ Level 1	Design structure changes are required	
	☑ MP ☐ ES	Bernoulli Type Single Robot		☐ Level 2	Simple part and work method change	

Before(Problems and Change Requests)

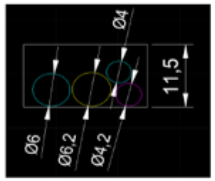
[AS]



用途	CABLE 型号	外径	半径	数量
PWR	RT-FLEX 2464 4C X 21AWG	5.4	33	1
ENC	RT-FLEX-SB 2464 2P X 25AWG	5.3	32	1
IO	RT-FLEX 2464 4C X 25AWG	3.8	23	1
	Air tube 4파이	4		1
	Air tube 6파이	6		1

After(Expected Effect)

[TO-BE]



用途	CABLE 型号	外径	半径	数量	备注
PWR	RT-FLEX 2464 4C X 23AWG	4.2	26	1	ITEM 1
ENC/IO	RT-FLEX-SB 2464 4P X 25AWG	6.2	38	1	ITEM 2
	Air tube 4파이	4		1	
	Air tube 6파이	6		1	

Due to th[illegible] solution on the Robots XS31-027 and JS29-013. (That is, the cable replacement solution)
We hope that NB can proceed with the hardware redesign of the Robot at the same time.

Related Division		Potential Issue	
☑	Mechanical design	☐	Instrument & tool additional cost incurred
☑	Electrical design	☐	Work process change and work time
☐	Manufacturing	☐	Mechanical failure
☐	Purchasing & Materials	☐	ETC :
☐	Quality		

2. Engineering Change Notice

적용 장비 타입 Applicable project type	Single Robot – BERNOULLI Type	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) □ 기타 (ETC)	ECR NO.	ECR-S-2605-03
		ECN NO.	ECN-E-2606-07
변경 상세 사유 Detailed reason for change	1. Flip Axis Chain 내 공간 확보 - Air Hose 파손 방지 1. Securing clearance inside the Flip Axis Chain to prevent Air Hose damage		

변경 사항 Changes

변경 전(Before)

용도	CABLE 규격	외경	곡률	수량
PWR	RT-FLEX 2464 4C X 21AWG	5.4	33	1
ENC	RT-FLEX-SB 2464 2P X 25AWG	5.3	32	1
IO	RT-FLEX 2464 4C X 25AWG	3.8	23	2
	Air tube 4파이	4		1
	Air tube 6파이	6		1

변경 후(After)

용도	CABLE 규격	외경	곡률	수량
PWR	RT-FLEX 2464 4C X 23AWG	4.2	26	1
ENC/IO	RT-FLEX-SB 2464 4P X 25AWG	6.2	38	1
	Air tube 4파이	4		1
	Air tube 6파이	6		1

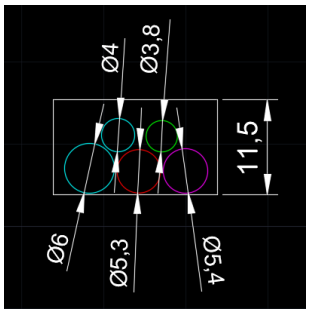
상세내용 Detail	- F Axis Motor Power : 4C x 21AWG - F Axis Motor Encoder : 2P x 25AWG - F Axis Sensor Cable : 4C x 25AWG	- F Axis Motor Power : 4C x 21AWG → **4C x 23AWG** - F Axis Motor Encoder & Sensor 통합 : 2P x 25AWG + 4C x 25AWG → **4P x 25AWG**

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2. Engineering Change Notice

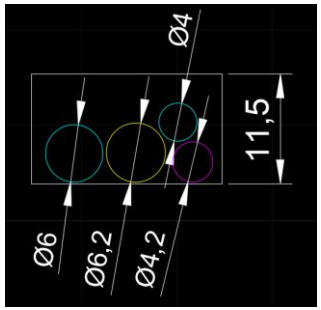
전체 이미지
Full Image

Before



용도	CABLE 규격	외경	곡률	수량
PWR	RT-FLEX 2464 4C X 21AWG	5.4	33	1
ENC	RT-FLEX-SB 2464 2P X 25AWG	5.3	32	1
IO	RT-FLEX 2464 4C X 25AWG	3.8	23	2
	Air tube 4파이	4		1
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After



용도	CABLE 규격	외경	곡률	수량	비고
PWR	RT-FLEX 2464 4C X 23AWG	4.2	26	1	ITEM 1
ENC/IO	RT-FLEX-SB 2464 4P X 25AWG	6.2	38	1	ITEM 2
	Air tube 4파이	4		1	
	Air tube 6파이	6		1	

검토 결과 Test Result	검토 여부 Test or not	Y / N	검토 부서 Department	전장팀	첨부파일 유무 Attachments	Y / N
승인 날짜 Approval date	1. Ninebell : 2026.06.16 2. Customer : / /		나인벨 승인 Ninebell approval	QC 2026.06.16 PASS	고객사 승인 Customer approval	**NB SALES Approve 06/23**
적용 일자 Effective date	2026.04.20		최초 적용 장비 Initially Applied Equipment	XS31-027		

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